Sourim Banerjee

✉ banerj47@purdue.edu ✉ sourim.banerjee10@gmail.com +1-765-409-3525 in sourim-banerjee10

Education

Purdue University August 2021 - Present
Ph.D. Student at School of Mechanical Engineering CGPA : 3.90/4

Indian Institute of Technology Kharagpur August 2016 - April 2021
B.Tech-M.Tech Dual Degree in Chemical Engineering CGPA : 9.21/10

Research Experiences

1. **Purdue University** August'21 - Present
   Advisor: Prof. Partha P. Mukherjee *Energy and Transport Science Lab(ETSL)*

   Doctoral thesis: Mesoscale Understanding of Instability at Solid-State Electrochemical Interfaces.

2. **Indian Institute of Technology Kharagpur** August'20 - April'21
   Advisor: Prof. Monojit Chakraborty
   CoAdvisor: Prof. Partha P. Mukherjee, Purdue University *Chemical Engineering Dept, IITkgp*

   Masters' thesis : Mesoscale Origin of Voids in Metal Electrodes using kinetic Monte Carlo method.

3. **Indian Institute of Technology Kharagpur** August'19 - April'20
   Advisor: Prof. Monojit Chakraborty
   CoAdvisor: Prof. Sunando Dasgupta *Chemical Engineering Dept, IITkgp*

   Undergraduate thesis: Metropolis Probability based kinetic Monte Carlo modeling of the protein nanoparticle drying pattern. (**Best B.Tech undergraduate thesis**)

4. **École polytechnique fédérale de Lausanne (EPFL)** May'19 - July'19
   Advisor: Prof.Francois Gallaire *Laboratory of Fluid Mechanics and Instabilities(LFMI), EPFL*

   Internship topic: Study of the dynamics of a confined bubble under gravity and Marangoni actuations using Finite Element Method in COMSOL Multiphysics software.

5. **Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)** May′18 - July′18
   Advisor: Prof.Ganesh Subramanian *Engineering Mechanics Unit (EMU), JNCASR*

   Internship topic: Study of the orientation dynamics of non-axisymmetric particle in various ambient planar linear flows.

Publications

1. **Sourim Banerjee**, Bairav S. Vishnugopi, Aditya Singla, Kautubh G. Naik and Partha P. Mukherjee."Void Formation and Evolution Dynamics for Lithium Metal and Solid Electrolyte Interfaces." ACS Applied Materials and Interfaces no. 18, (2026): 7981-7991.
2. **Sourim Banerjee**, Bairav S. Vishnugopi, and Partha P. Mukherjee. "Probing the Impact of Vacancy Diffusion on Void Dynamics at the Lithium Metal-Solid Electrolyte Interface."Advanced Science (2026): e15827.
3. **Sourim Banerjee**, Bairav S. Vishnugopi, Guang Lin and Partha P. Mukherjee. "Mechanistic Insights into Instability from Single Void Dynamics at the Lithium Metal Solid-Electrolyte Interface." (*accepted*)
4. **Sourim Banerjee**, Ahmad Ahmad, Bairav S. Vishnugopi and Partha P. Mukherjee. "Impact of local stress on the growth of a single void during stripping in Li metal solid-state batteries - A coupled kMC-FFT approach" (*in preparation*)
5. **Sourim Banerjee**, Ahmad Ahmad, Bairav S. Vishnugopi and Partha P. Mukherjee. "A kinetic Monte Carlo study on the Impact of Creep on Interfacial Degradation in Lithium Metal Anode Solid-State Batteries" (*in preparation*)
6. **Sourim Banerjee**, Bairav S. Vishnugopi and Partha P. Mukherjee. "Investigating the Role of Secondary Metal Phase in Influencing Interfacial Stability in Lithium Solid-State Batteries" (*in preparation*)

7. Min-Gi Jeong, Kelsey B. Hatzell, **Sourim Banerjee**, Bairav S. Vishnugopi, and Partha P. Mukherjee. "Temperature Impact on Lithium Metal Morphology in Lithium Reservoir-Free Solid-State Batteries." PRX Energy 3, no. 2 (2024): 023003.

8. Debanjali Chatterjee, Aditya Singla, **Sourim Banerjee**, Anuththara Alujjage, Pretty Mitra and Partha P. Mukherjee. "MoChA: Modeling, Characterization & Analytics in Electrochemical Energy Systems." ACS Energy Letters(2025): 3430-3436.

Conferences

1. **Sourim Banerjee**, and Partha P. Mukherjee. "Mechanistic Insights into Single-Void Growth and Morphology In Lithium Metal Solid-State Batteries." In 2026 AIChE Annual Meeting in Minneapolis, MN.

2. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, and Partha P. Mukherjee. "Influence of Surface Diffusion Interactions on Void Formation in Solid-State Batteries." In 248th Electrochemical Society Meeting Abstracts, October, 2025.

3. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, and Partha P. Mukherjee. "Role of Surface Diffusion Mechanism in Void Formation and Instability at Solid/Solid Electrochemical Interfaces." In Materials Research Society Abstracts, Spring 2025.

4. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, Kaustubh Girish Naik, and Partha P. Mukherjee. "Mechanism of Void Formation in Lithium Metal Solid-State Batteries." In Electrochemical Society Meeting Abstracts 242, no. 4, pp. 383-383. The Electrochemical Society, Inc., 2022.

Computational Proficiency

Programming Languages	Matlab • Python • C++
Computational Methods and Solvers	Kinetic Monte Carlo • FEniCS (FEM) • FFT Micromechanics solver
Software Skills	GeoDict • OVITO • LAMMPS • LaTeX• COMSOL Multiphysics • ASPEN • Solid Works

Courses Taken

Core courses: Fluid mechanics, Heat and Mass transfer, Reaction Engineering, Thermodynamics, Transport Phenomena, Process Modeling and Simulation, Instrumentation and Process Control, and Microfluidics.

Relevant Courses: Fundamentals of Electrochemical Energy Systems, Materials Engineering Fundamentals, Introduction to Scientific Machine Learning, Calculus, Engineering Mathematics, Regression and Time Series model, Probability and Stochastic Process.

Teaching Experiences

1. **Fundamentals of Electrochemical Energy Systems**(Instructor: Prof. Partha P. Mukherjee) Spring 2024
   - Developed and refined lecture materials, homework assignments, and examinations to strengthen course content and student learning, and independently instructed a full class lecture.
   - Mentored 32 graduate students in designing and solving practical electrochemical energy storage problems as part of the course project, resulting in 2 peer-reviewed journal publications.

2. **Undergraduate Heat and Mass Transfer**(Instructor: Prof. Thomas Beechem) Fall 2024
   - Provided individualized academic support through office hours and assessed examinations and homework assignments, delivering constructive feedback to enhance student learning.

Academic Achievements

- Selected for the prestigious **Purdue Trailblazers in Engineering** in 2026 as one of the top 25 fellows from over 200 applicants nationwide. This workshop prepares outstanding Ph.D. students and postdoctoral researchers for successful careers in engineering academia.
- Awarded **Graduate Student Award for Outstanding Service Scholarship** in 2024 in recognition of exceptional contributions to strengthening Purdue's scientific and research student community through leadership and community building.

- Awarded **Arrasmith Fellowship** with **Lozer Assistantship** in 2021 for extraordinary graduate school application and exceptional academic potential at Purdue University.
- **Department Rank 4** among 42 B.Tech-M.Tech Dual degree students in Chemical Engineering Department at IIT Kharagpur.
- Secured All India Rank **379** in prestigious KVPY (Kishore Vaigyanik Protsahan Yajana- translated to **Young Scientist Encourage Scheme**) scholarship and awarded prestigious JBNSTS (Jagadis Bose National Science Talent Search) scholarship, 2016.

Service & Mentorship

- **Secretary & President, Electrochemical Society Student Chapter Purdue** August'21 - January '25
  - Organized professional seminars, workshops, and panel discussions connecting 50+ students, researchers, and faculty with leading experts to promote professional development and interdisciplinary collaboration. Led hands-on electrochemical demonstrations for 20+ high school students at Nano Expo Day (April 2023), fostering interest in STEM and scientific research at Purdue.
  - Served as Chapter President (Aug. 2024–Jan. 2025) leading the organization of the Women in Sustainability event and the chapter's participation in the Fall 2024 Research Expo hosted by the Purdue Institute for a Sustainable Future.
- **Treasurer, Engineering Academic Career Club** August'23 - August'24
  - Managed organization finances by overseeing organizational funds, developing annual budgets, and securing external funding through a successful grant application to support chapter initiatives.
  - Organized professional panel discussion on future faculty applications and post-Ph.D. career pathways in April 2024, equipping graduate students and postdoctoral researchers with essential career development resources and guidance.
- **Mentor, IIT Kharagpur, Purdue University**
  - Mentored 6 first-year Chemical Engineering students by providing guidance on academics, extracurricular involvement, and campus life as part of the Student Welfare Group (SWG) in 2018-2019 academic year.
  - Mentored junior graduate students by providing guidance on coursework, research planning, problem solving, and the development of a strong foundation in fundamental concepts through regular discussions and brainstorming during PhD.